

STATE VARIABLE ANALYSIS

UNIT I

COMPUTER AND SYSTEMS ENGINEERING DEPARTMENT

INTRODUCTION

- The classical control theory and methods that we have been using in class are based on a simple input-output description of the plant, usually expressed as a transfer function. These methods do not use any knowledge of the interior structure of the plant, and limit us to single-input single-output (SISO) systems, and as we have seen allows only limited control of the closed-loop behavior when feedback control is used.
- Modern control theory solves many of the limitations by using a much “richer” description of the plant dynamics. The so-called state-space description provide the dynamics as a set of coupled first-order differential equations in a set of internal variables known as *state variables*, together with a set of algebraic equations that combine the state variables into physical output variables.

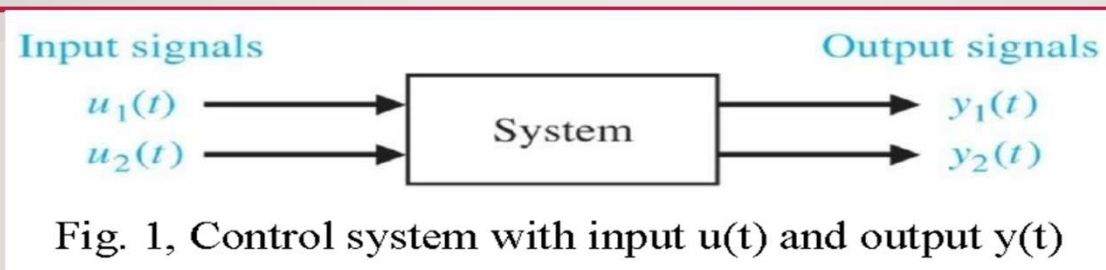
STATE-VARIABLE APPROACH (DEFINITIONS)

- **State:** The concept of the *state* of a dynamic system refers to a minimum set of variables, known as *state variables*, that fully describe the system and its response to any given set of inputs.
- In particular a *state-determined* system model has the characteristic that:
 - A mathematical description of the system in terms of a minimum set of variables $x_i(t)$, $i = 1, \dots, n$,
 - together with knowledge of those variables at an initial time t_0
 - and the system inputs for time $t \geq t_0$,
 - ➔ are sufficient to predict the future system state and outputs for all time $t > t_0$.
 - ➔ Note that, in dealing with linear time-invariant systems, we usually choose the reference time t_0 to be zero.

STATE-VARIABLE APPROACH (DEFINITIONS)

- **State Variables:** are the smallest set of variables needed to fully describe the dynamic system.
- **State Vector:** if there are n state variables $\{x_1(t), x_2(t), \dots, x_n(t)\}$ that are needed to completely describe the behavior of a given system, then these n state variables can be considered as the n components of a vector $x(t)$. such a vector is called a state vector.
- **State Space:** the n -dimensional space whose coordinate axes consist of the x_1 axis, x_2 axis, ..., x_n axis is called a state space. Any state can be represented by a point in the state space.

1- STATE, OUTPUT AND DYNAMIC EQUATIONS



- Assume we have the set of n state variables $(x_1, x_2, \dots, x_n)$ that can describe the above system. Therefore, the relation that govern the derivative of the state variable with the state variables and the system input is called the **state equation**.

$$\dot{x}_1 = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n + b_{11}u_1 + \dots + b_{1m}u_m$$

$$\dot{x}_2 = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n + b_{21}u_1 + \dots + b_{2m}u_m$$

$$\vdots$$
$$\vdots$$

$$\dot{x}_n = a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n + b_{n1}u_1 + \dots + b_{nm}u_m$$

1- STATE, OUTPUT AND DYNAMIC EQUATIONS

- The above equations can be arranged in matrix form as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \dots & \dots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \dots & \dots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix}$$

Or

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (1)$$

Diagram illustrating the matrix dimensions for the state equation:

- $\dot{x}(t)$ is $n \times 1$.
- A is $n \times n$.
- $x(t)$ is $n \times 1$.
- B is $n \times m$.
- $u(t)$ is $m \times 1$.

the state vector: $x(t)$
system matrix : A
input matrix : B
input vector : $u(t)$

1- STATE, OUTPUT AND DYNAMIC EQUATIONS

- On the other hand, the relation that govern the system output with the state variables and the system input is called the **output equation**.

$$\begin{aligned} y_1 &= c_{11}x_1 + c_{12}x_2 + \cdots + c_{1n}x_n + d_{11}u_1 + \cdots + d_{1m}u_m \\ y_2 &= c_{21}x_1 + c_{22}x_2 + \cdots + c_{2n}x_n + d_{21}u_1 + \cdots + d_{2m}u_m \\ &\vdots \\ y_k &= c_{k1}x_1 + c_{k2}x_2 + \cdots + c_{kn}x_n + d_{k1}u_1 + \cdots + d_{km}u_m \end{aligned}$$

Or

$$y(t) = Cx(t) + Du(t) \quad (2)$$

the state vector : $x(t)$
 system matrix : C
 system direct-link, feed-forward matrix : D
 output vector : $y(t)$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_k \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ c_{k1} & c_{k2} & \cdots & c_{kn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \vdots & \vdots & \cdots & \vdots \\ d_{k1} & d_{k2} & \cdots & d_{km} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix}$$

Diagram illustrating the dimensions of the matrices and vectors in the output equation (2):

- $y(t)$ is a $k \times 1$ vector.
- C is a $k \times n$ matrix.
- $x(t)$ is an $n \times 1$ vector.
- D is a $k \times m$ matrix.
- $u(t)$ is an $m \times 1$ vector.

1- STATE, OUTPUT AND DYNAMIC EQUATIONS

- Both of the state equation given by (1) and the output equation given by (2) together are called the **dynamic equation**.

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

- The choice of state variables for a given system is not unique.
- The requirement in choosing the state variables is that they be linearly independent and that a minimum number of them be chosen.

1- STATE, OUTPUT AND DYNAMIC EQUATIONS

- The dynamic equation given above can be expressed by vector block diagram representation shown in Fig. 2. This general block diagram shows the matrix operations from input to output in terms of the **A**, **B**, **C**, **D** matrices, but does not show the path of individual variables.

Matrix **A** is called the system matrix,
Matrix **B** is called the input matrix,
Matrix **C** is called the output matrix,
Matrix **D** is called the feedforward matrix.

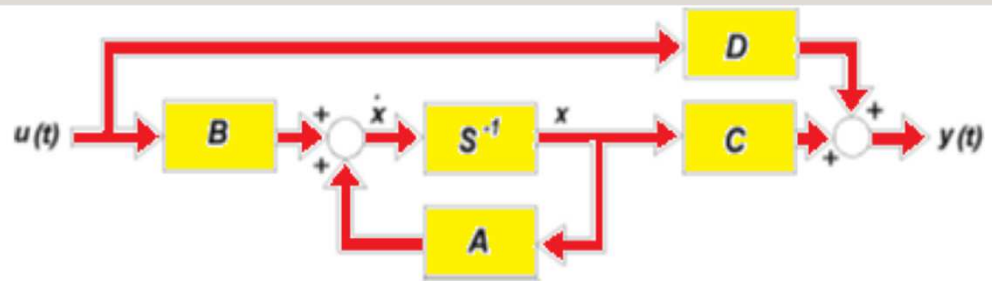


Fig. 2, Vector block diagram for linear system described by dynamic equation

REPRESENTING AN ELECTRICAL NETWORK

- Drive the dynamic equation for the electric circuit shown in Fig. 3.
- First, assume the state variables are $x_1 = v_c$ and $x_2 = i_L$, and the output $y = v_o$.

$$u(t) = i_L + i_c \quad i_c = C \frac{dv_c}{dt} = u(t) - i_L \quad \longrightarrow \quad \frac{dv_c}{dt} = \frac{1}{C} u(t) - \frac{1}{C} i_L$$

$$L \frac{di_L}{dt} + R i_L = v_c \quad \longrightarrow \quad \frac{di_L}{dt} = \frac{1}{L} v_c - \frac{R}{L} i_L$$

$$v_o = R i_L \quad \longrightarrow \quad y = R i_L$$

$$\dot{x}_1(t) = \frac{1}{C} u(t) - \frac{1}{C} x_2$$

$$\dot{x}_2(t) = \frac{1}{L} x_1 - \frac{R}{L} x_2$$

$$y(t) = R x_2$$

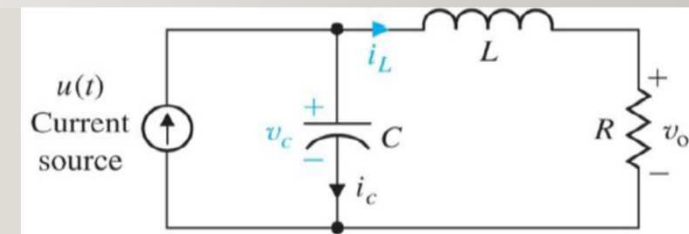


Fig. 3, RLC circuit

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{C} \\ \frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{C} \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & R \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

CONVERTING A TRANSFER FUNCTION TO STATE SPACE (DECOMPOSITION)

- One advantage of the state-space representation is that it can be used for the simulation of physical systems on the digital computer.
- Thus, if we want to simulate a system that is represented by a transfer function, we must first convert the transfer function representation to state space.
- At first, we select a set of state variables, called phase variables, where each subsequent state variable is defined to be the derivative of the previous state variable.
 1. Converting a Transfer Function with Constant Term in Numerator
 2. Converting a Transfer Function with Polynomial in Numerator

CONVERTING A TRANSFER FUNCTION WITH CONSTANT TERM IN NUMERATOR

- **Example:** Find the state-space representation in phase-variable form for the transfer function given below:

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

Solution:

$$(s^3 + 9s^2 + 26s + 24)C(s) = 24R(s)$$

inverse Laplace transform, assuming zero initial conditions $\Rightarrow$

$$\frac{d^3c(t)}{dt^3} + 9\frac{d^2c(t)}{dt^2} + 26\frac{dc(t)}{dt} + 24c(t) = 24r(t)$$

CONVERTING A TRANSFER FUNCTION WITH CONSTANT TERM IN NUMERATOR

$$\frac{d^3c(t)}{dt^3} + 9\frac{d^2c(t)}{dt^2} + 26\frac{dc(t)}{dt} + 24c(t) = 24r(t)$$

$$\text{Let } x_1(t) = c(t) \rightarrow \dot{x}_1(t) = \frac{dc(t)}{dt} \rightarrow \dot{x}_1(t) = x_2(t)$$

$$\text{Also, } x_2(t) = \frac{dc(t)}{dt} \rightarrow \dot{x}_2(t) = \frac{d^2c(t)}{dt^2} \rightarrow \dot{x}_2(t) = x_3(t)$$

$$\text{Also, } x_3(t) = \frac{d^2c(t)}{dt^2} \rightarrow \dot{x}_3(t) = \frac{d^3c(t)}{dt^3} \rightarrow \dot{x}_3(t) = 24r(t) - 24x_1(t) - 26x_2(t) - 9x_3(t)$$

- $\dot{x}_1 = 0x_1 + 1x_2 + 0x_3 + 0r$

- $\dot{x}_2 = 0x_1 + 0x_2 + 1x_3 + 0r$

- $\dot{x}_3 = -24x_1 + -26x_2 + -9x_3 + 24r$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r(t)$$

CONVERTING A TRANSFER FUNCTION WITH CONSTANT TERM IN NUMERATOR

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r(t)$$

- Notice that the third row of the system matrix has the same coefficients as the denominator of the transfer function but negative and in reverse order. The matrix A obtained is with dimension 3×3 and matrix B obtained is with dimension 3×1 are in phase variable canonical form.
- The output equation:

$$c(t) = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + [0] r(t)$$

CONVERTING A TRANSFER FUNCTION WITH CONSTANT TERM IN NUMERATOR

$$\dot{x}_1(t) = x_2(t)$$

$$\rightarrow \dot{x}_2(t) = x_3(t)$$

$$\dot{x}_3(t) = 24r(t) - 24x_1(t) - 26x_2(t) - 9x_3(t)$$

- The above 3 equations can be represented by the equivalent block diagram, shown in Fig. 7, that represents the phase variables.

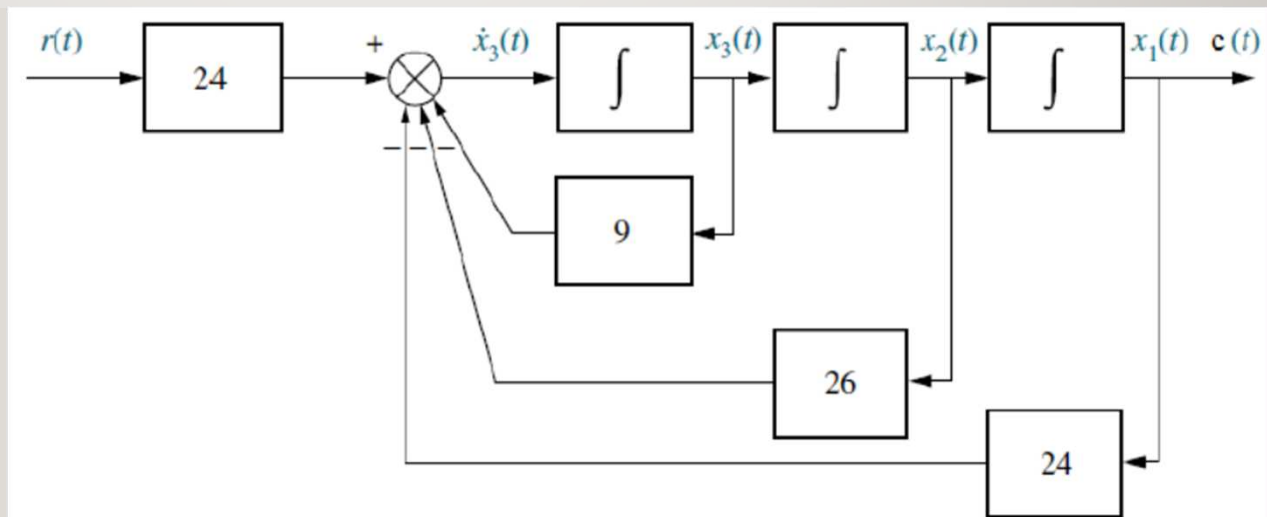


Fig. 7, Block diagram shows the phase variables

CONVERTING A TRANSFER FUNCTION WITH POLYNOMIAL IN NUMERATOR

- There are three types of decomposition:
 1. Direct decomposition,
 2. Cascaded decomposition (self study)
 3. and Parallel decomposition (self study)

1 DIRECT DECOMPOSITION

- Example: convert the given transfer function to state space representation

$$\frac{Y(s)}{U(s)} = \frac{10(S + 1)}{(S + 2)^2(S + 5)}$$

- Both numerator and denominator of the T.F. must be in polynomial format as:

$$\frac{Y(s)}{U(s)} = 10 \frac{S + 1}{S^3 + 9S^2 + 24S + 20}$$

- **Step #1** :All the power of S must be negative; this can be achieved by dividing by S^3

$$\frac{Y(s)}{U(s)} = \frac{S^{-2} + S^{-3}}{1 + 9S^{-1} + 24S^{-2} + 20S^{-3}}$$

1 DIRECT DECOMPOSITION

- **Step #2** : Multiply both the numerator and denominator by dummy variable $X(S)$

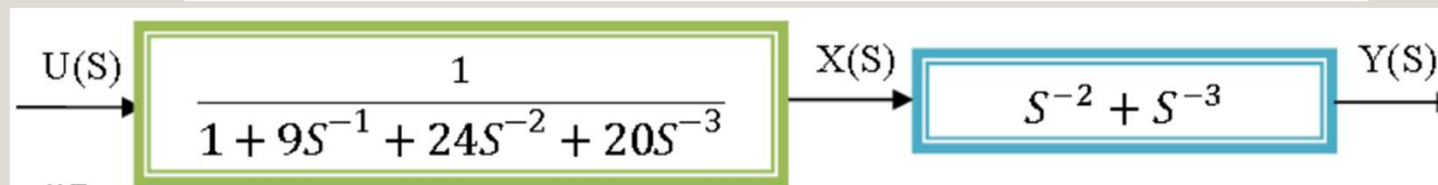
$$\frac{Y(s)}{U(s)} = \frac{S^{-2} + S^{-3}}{1 + 9S^{-1} + 24S^{-2} + 20S^{-3}} \times \frac{X(S)}{X(S)}$$

- **Step #3** Equating numerators

$$Y(S) = (S^{-2} + S^{-3}) X(S)$$
$$Y(S) = S^{-2}X(S) + S^{-3}X(S) \quad (1)$$

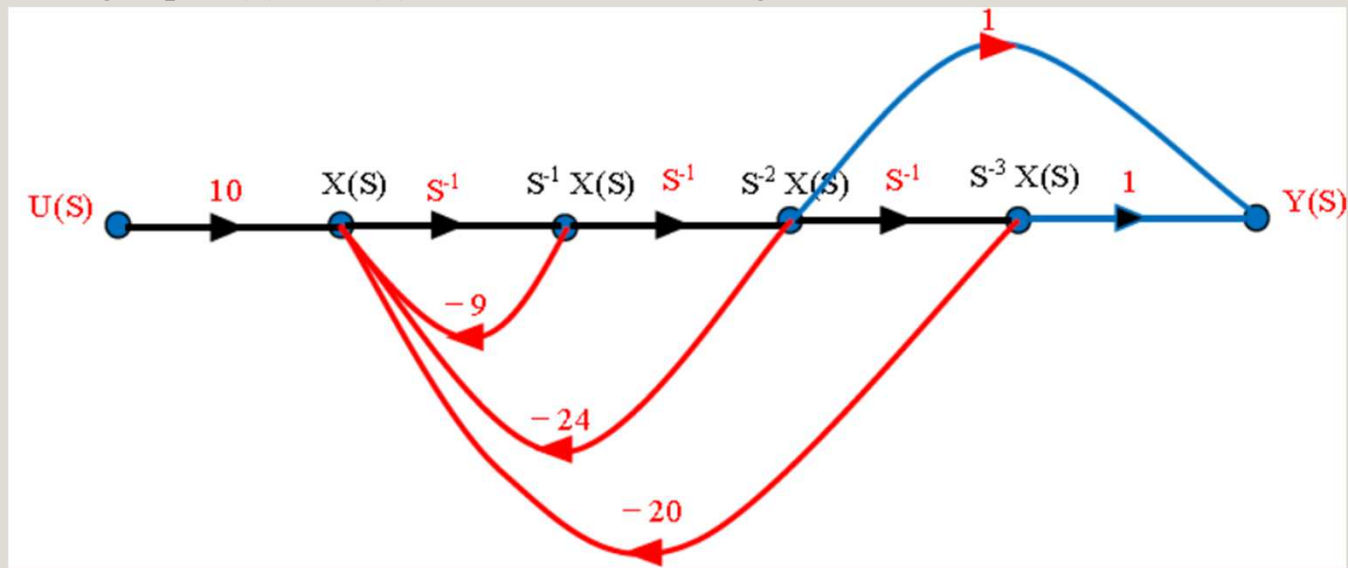
- **Step #4** Equating denominators

$$U(S) = (1 + 9S^{-1} + 24S^{-2} + 20S^{-3}) X(S)$$
$$X(S) = U(S) - 9S^{-1}X(S) - 24S^{-2}X(S) - 20S^{-3}X(S) \quad (2)$$



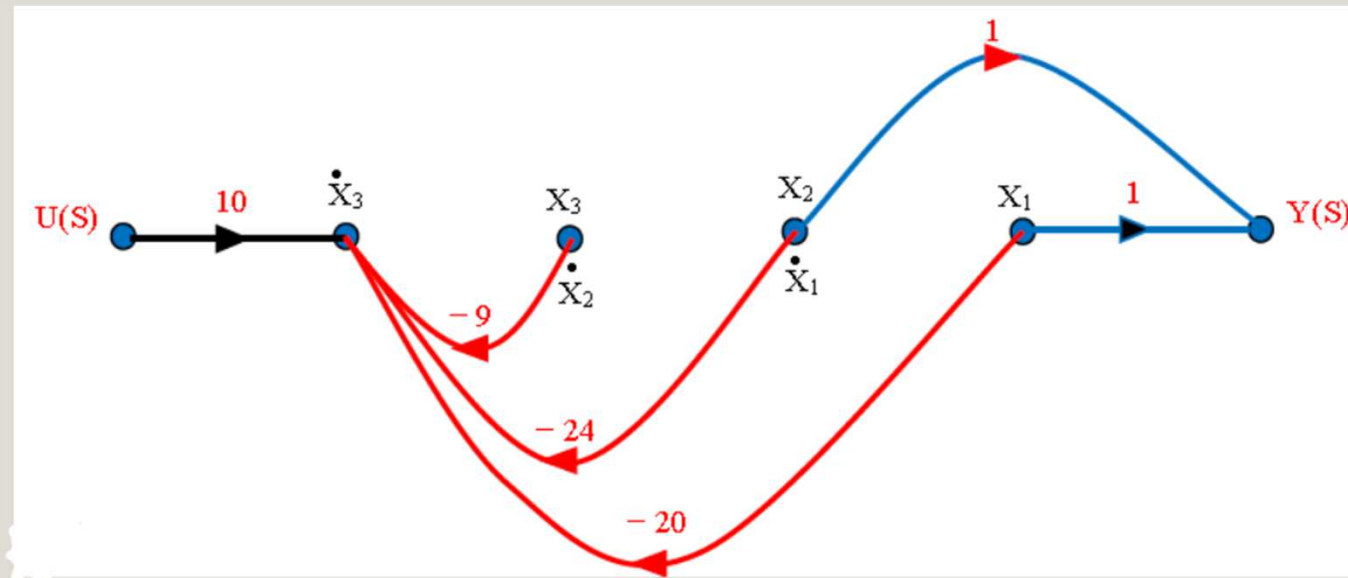
1 DIRECT DECOMPOSITION

- **Step #5** : Using Eqns. (1) and (2) to draw the state diagram as shown below:



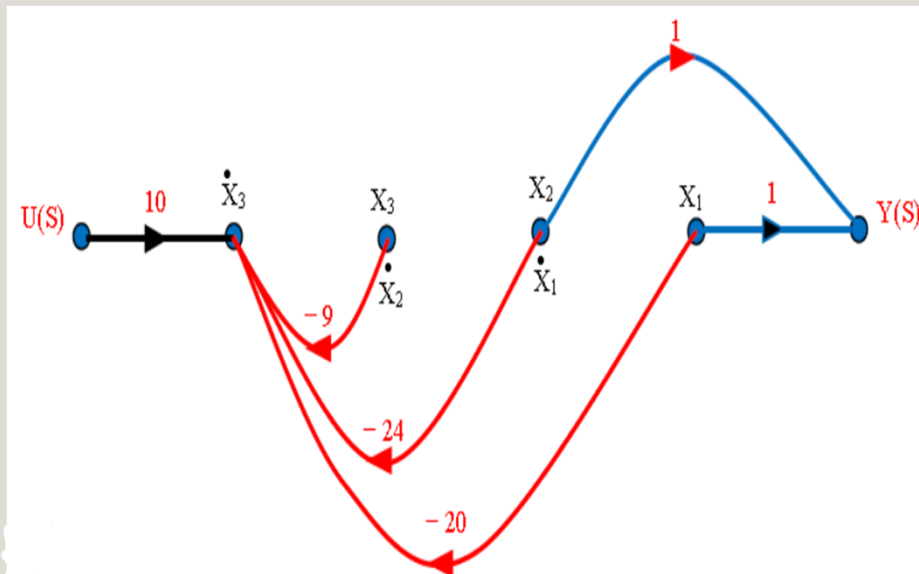
1 DIRECT DECOMPOSITION

- **Step #6 :** Rename the state variables (after each integrator) from right to left, and remove the integrator S^{-1} as shown in the state diagram given below.



1 DIRECT DECOMPOSITION

- **Step #7 :** Write the state and output equation [Dynamic Equation] as follows:



$$\dot{x}_1(t) = x_2$$

$$\dot{x}_2(t) = x_3$$

$$\dot{x}_3(t) = 10u - 9x_3 - 24x_2 - 20x_1$$

$$y = x_1 + x_2$$



$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -20 & -24 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u(t)$$

$$y(t) = [1 \quad 1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + [0] u(t)$$

CONVERTING FROM STATE SPACE TO A TRANSFER FUNCTION

Idea

- Assume a linear control system with an input vector $u(t)$ and described by the state eqn.

$$\dot{x}(t) = Ax(t) + Bu(t)$$

- Taking Laplace transform

$$sX(s) - x(0) = AX(s) + B U(s)$$

$$X(s)[sI - A] = x(0) + B U(s)$$

- In order to obtain the transfer function, the initial values $x(0)$ must be zero, then

$$X(s) = [sI - A]^{-1}B U(s) \quad (4)$$

CONVERTING FROM STATE SPACE TO A TRANSFER FUNCTION

- The output equation is $y(t) = Cx(t) + Du(t)$
- Taking Laplace transform $Y(S) = CX(S) + DU(S)$ (5)
- Substituting by the value of $X(S)$ given by (4) into the output equation (5), then

$$\begin{aligned} Y(S) &= C\{[sI - A]^{-1}B U(s)\} + DU(S) \\ Y(S) &= \{C [sI - A]^{-1}B + D\} U(S) \\ T.F.G(S) &= \frac{Y(S)}{U(S)} = C [sI - A]^{-1}B + D \quad (6) \end{aligned}$$

CONVERTING FROM STATE SPACE TO A TRANSFER FUNCTION

- **Example:** Consider the dynamic equation for the system with output $y(t)$ and input $u(t)$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

- Determine the system transfer function ?
- First, we must calculate $[sI - A]^{-1}$.

$$[sI - A]^{-1} = \begin{bmatrix} \frac{s+3}{(s+2)(s+1)} & \frac{1}{(s+2)(s+1)} \\ \frac{-2}{(s+2)(s+1)} & \frac{s}{(s+2)(s+1)} \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$
$$A^{-1} = (1/|A|) \times \text{Adj } A$$
$$= [1/(ad - bc)] \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

CONVERTING FROM STATE SPACE TO A TRANSFER FUNCTION

$$\frac{Y(S)}{U(S)} = C [sI - A]^{-1} B + D$$

$$\frac{Y(S)}{U(S)} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{S+3}{(S+2)(S+1)} & \frac{1}{(S+2)(S+1)} \\ \frac{-2}{(S+2)(S+1)} & \frac{S}{(S+2)(S+1)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix}$$

$$\frac{Y(S)}{U(S)} = \begin{bmatrix} \frac{-2}{(S+2)(S+1)} & \frac{S}{(S+2)(S+1)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{S}{(S+2)(S+1)}$$

Use Matlab code to convert the 4 matrices to the required T.F.

```
%% A = [0 1; -2 -3];  
%% B = [0; 1];  
%% C = [0 1];  
%% D = [0];  
%% SYS = ss(A B C D);  
%% T = tf(SYS)
```

$$T = \frac{s}{s^2 + 3s + 2}$$

CHARACTERISTIC EQUATION

- The characteristic equation plays an important role in the study of linear systems. It can be defined with respect to the differential equation, the transfer function, or state equations.

1- Characteristic Equation from Differential Equation

In T.F. all initial values must be zero, therefore, if we have certain D.E.

$$\begin{aligned}\ddot{c}(t) + 9\dot{c}(t) + 24c(t) &= 10\dot{r}(t) + 10r(t) \\ S^3C(s) + 9S^2C(s) + 24SC(s) + 20C(s) &= 10SR(s) + 10R(s) \\ C(s)\{S^3 + 9S^2 + 24S + 20\} &= R(s)\{10S + 10\}\end{aligned}$$

Then the characteristic equation is

$$S^3 + 9S^2 + 24S + 20 = 0$$

CHARACTERISTIC EQUATION

2 Characteristic Equation from Transfer Function

- If the transfer function of a control system is given, the characteristic equation is obtained by equating the denominator polynomial of the T.F. to zero.

- Suppose the transfer function of control system is given as:

$$\frac{C(s)}{R(s)} = \frac{10(S + 1)}{(S + 2)^2(S + 5)}$$

- The T.F. must be in polynomial format as:

$$\frac{C(s)}{R(s)} = \frac{10S + 10}{S^3 + 9S^2 + 24S + 20}$$

- Then, the characteristic equation is:

$$S^3 + 9S^2 + 24S + 20 = 0$$

CHARACTERISTIC EQUATION

3 Characteristic Equation from State Equation

The T.F. obtained from the matrices A, B, C, and D as:

$$\begin{aligned}\frac{Y(S)}{U(S)} &= C [sI - A]^{-1} B + D \\ \frac{Y(S)}{U(S)} &= \frac{C \operatorname{adj} [sI - A] B}{|sI - A|} + D \\ \frac{Y(S)}{U(S)} &= \frac{C \operatorname{adj} [sI - A] B + D |sI - A|}{|sI - A|}\end{aligned}$$

As we said before, the characteristic equation is obtained by equating the denominator polynomial of the T.F. to zero, then it can be obtained as:

$$|sI - A| = 0$$

CHARACTERISTIC EQUATION

- Example suppose the matrix A is:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -4 & 2 \\ -1 & -5 & -3 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & -4 & 2 \\ -1 & -5 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s+4 & -2 \\ 1 & 5 & s+3 \end{bmatrix}$$

- the characteristic equation is:

$$|sI - A| = s\{(s+4)(s+3) + 10\} + 2 = s^3 + 7s^2 + 22s + 2 = 0$$

TIME DOMAIN SOLUTION OF STATE EQUATIONS

- Assume a linear control system with an input vector $u(t)$ and described by the state equation:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$\dot{x}(t) - Ax(t) = Bu(t)$$

- Multiplying both sides by an exponential function e^{-At}

$$e^{-At}\{\dot{x}(t) - Ax(t)\} = e^{-At}\{Bu(t)\} \quad \text{Eq (1)}$$

- Note that the left hand side can be rewritten as:

$$e^{-At}\{\dot{x}(t) - Ax(t)\} = \frac{d}{dt}\{e^{-At}x(t)\}$$

- Replace the left hand side in Eq (1)

$$\frac{d}{dt}\{e^{-At}x(t)\} = e^{-At}\{Bu(t)\}$$

TIME DOMAIN SOLUTION OF STATE EQUATIONS

- Integrating both sides in the interval from 0 to t

$$e^{-At}x(t)|_0^t = \int_0^t e^{-A\tau}\{Bu(\tau)\}.d\tau$$

$$e^{-At}x(t) - x(0) = \int_0^t e^{-A\tau}Bu(\tau).d\tau$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau).dt$$

- the first term represents the unforced response and the second term represents the forced response.

TIME DOMAIN SOLUTION OF STATE EQUATIONS

- We can handle the solution of state equation from Laplace point of view by taking Laplace transform for both sides of the state equation, then:

$$sX(s) - x(0) = AX(s) + B U(s)$$

$$[sI - A]X(s) = x(0) + B U(s)$$

$$X(s) = [sI - A]^{-1}x(0) + [sI - A]^{-1}B U(s)$$

- By comparing the obtained solution here with that obtained previously, we find the state transition matrix

$$\Phi(S) = [sI - A]^{-1} \text{ and } \Phi(t) = \int^{-1} [sI - A]^{-1}$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau).dt$$

TIME DOMAIN SOLUTION OF STATE EQUATIONS

- Some properties of the state transition matrix $\Phi(t)$

$$1) \Phi(t) = e^{At} = \int^{-1} [sI - A]^{-1}$$

$$2) \Phi(0) = I$$

$$3) \Phi^{-1}(t) = \Phi(-t)$$

$$4) \Phi(t_1 + t_2) = \Phi(t_1) \times \Phi(t_2) = \Phi(t_2) \times \Phi(t_1)$$

- To get $x(t)$, this is done simply by taking inverse Laplace transform for both sides:

*zero-input
response*

$$x(t) = L^{-1}\{[sI - A]^{-1}\{x(0) + B U(s)\}\}$$

*zero-state
response,*

$$x(t) = L^{-1}\{\varphi(s)\} x(0) + L^{-1}\{\varphi(s) B U(s)\} \quad (1)$$

- Above equation is called the state transition equation and the term $L^{-1}\{[sI - A]^{-1}\}$ is called the state transition matrix $\varphi(t)$

EXAMPLE

Consider the state equation shown below.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$
$$y = [1 \quad 3] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Determine the state transition equation $x(t)$ for a unit step input and $x_1(0)=1$, $x_2(0)=2$

$$[sI - A] = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$\Phi(s) = [sI - A]^{-1} = \frac{1}{\Delta(s)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$\Delta(s) = \det(sI - A) = s(s+3) + 2 = s^2 + 3s + 2 = (s+2)(s+1)$$

$$\Phi(s) = \begin{bmatrix} \frac{s+3}{(s+2)(s+1)} & \frac{1}{(s+2)(s+1)} \\ \frac{-2}{(s+2)(s+1)} & \frac{s}{(s+2)(s+1)} \end{bmatrix}$$

EXAMPLE

To get the state transition matrix $\varphi(t)$, we use partial fraction of each term in $\varphi(S)$

$$\Phi(s) = \begin{bmatrix} \frac{-1}{(S+2)} + \frac{2}{(S+1)} & \frac{-1}{(S+2)} + \frac{1}{(S+1)} \\ \frac{2}{(S+2)} - \frac{2}{(S+1)} & \frac{2}{(S+2)} + \frac{-1}{(S+1)} \end{bmatrix}$$

$$\varphi(t) = \begin{bmatrix} -e^{-2t} + 2e^{-t} & -e^{-2t} + e^{-t} \\ 2e^{-2t} - 2e^{-t} & 2e^{-2t} - e^{-t} \end{bmatrix}$$

$$\varphi(S)B U(s) = \begin{bmatrix} \frac{S+3}{(S+2)(S+1)} & \frac{1}{(S+2)(S+1)} \\ \frac{-2}{(S+2)(S+1)} & \frac{S}{(S+2)(S+1)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{S}$$

EXAMPLE

$$\varphi(S) B U(s) = \left[\frac{\frac{1}{S(S+2)(S+1)}}{\frac{S}{S(S+2)(S+1)}} \right] = \left[\begin{array}{c} 0.5 + \frac{0.5}{S+2} + \frac{-1}{S+1} \\ \frac{-1}{S+2} + \frac{1}{S+1} \end{array} \right]$$

$$L^{-1}\{\varphi(S) B U(s)\} = \left[\begin{array}{c} 0.5 + 0.5 e^{-2t} - e^{-t} \\ -e^{-2t} + e^{-t} \end{array} \right]$$

$$x(t) = \left[\begin{array}{cc} -e^{-2t} + 2e^{-t} & -e^{-2t} + e^{-t} \\ 2e^{-2t} - 2e^{-t} & 2e^{-2t} - e^{-t} \end{array} \right] \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \left[\begin{array}{c} 0.5 + 0.5 e^{-2t} - e^{-t} \\ -e^{-2t} + e^{-t} \end{array} \right]$$

$$x(t) = \left[\begin{array}{c} -3e^{-2t} + 4e^{-t} \\ 6e^{-2t} - 4e^{-t} \end{array} \right] + \left[\begin{array}{c} 0.5 + 0.5 e^{-2t} - e^{-t} \\ -e^{-2t} + e^{-t} \end{array} \right] = \left[\begin{array}{c} 0.5 - 2.5e^{-2t} + 3e^{-t} \\ 5e^{-2t} - 3e^{-t} \end{array} \right]$$

On the other hand, we can calculate the output $y(t)$ based on the output equation

$$y(t) = [1 \quad 3] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = [1 \quad 3] \left[\begin{array}{c} 0.5 - 2.5e^{-2t} + 3e^{-t} \\ 5e^{-2t} - 3e^{-t} \end{array} \right] = 0.5 + 12.5e^{-2t} - 6e^{-t}$$